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Variability in Causal Effects and Noncompliance in a Multisite Trial: A Bivariate Hierarchical Generalized Random Coefficients Model for a Binary Outcome

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ABSTRACT

Within each of 170 physicians, patients were randomized to access e-assist, an online program that aimed to increase colorectal cancer screening (CRCS), or control. Compliance was partial: 78.34% of the experimental patients accessed e-assist while no controls were provided the access. Of interest are the average causal effect of assignment to treatment and the complier average causal effect as well as the variation of these causal effects across physicians. Each physician generates probabilities of screening for experimental compliers (experimental patients who accessed e-assist), control compliers (controls who would have accessed e-assist had they been assigned to e-assist), and never takers (patients who would have avoided e-assist no matter what). Estimating physician-specific probabilities jointly over physicians poses novel challenges. We address these challenges by maximum likelihood, factoring a "complete-data likelihood" uniquely into the conditional distribution of screening and partially observed compliance given random effects and the distribution of random effects. We marginalize this likelihood using adaptive Gauss–Hermite quadrature. The approach is doubly iterative in that the conditional distribution defies analytic evaluation. Because the small sample size per physician constrains estimability of multiple random effects, we reduce their dimensionality using a shared random effects model having a factor analytic structure. We assess estimators and recommend sample sizes to produce reasonably accurate and precise estimates by simulation, and analyze data from a trial of a CRCS intervention.

1 | Introduction

Colorectal cancer screening (CRCS) is underused compared to other cancer screenings [1]. While 93% of the insured due for CRCS receive a recommendation for screening when visiting a physician office, only 54% complete screening in the following year [2]. To improve the screening rate, a team of investigators developed an online CRCS decision support program called "e-assist: Colon Health" or "e-assist" that reinforces benefits, facilitates informed decision making, and addresses common

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barriers to and questions regarding CRCS following a physician recommendation for screening [3].

To test the effectiveness of e-assist, a multisite randomized controlled trial (MRCT) was conducted at primary care clinics located throughout the city of Detroit and the surrounding suburban tricounty area. Eligible patients aged 50–75 due for CRCS and having online patient portal accounts were randomly assigned to e-assist or control within each of 170 physicians and emailed to access it by secure patient portals. Those assigned to control were routed to publicly available online colorectal cancer and screening education material, and not allowed to access e-assist. The outcome variable is equal to 1 if a patient completes screening and 0 otherwise within one year after receiving a physician recommendation for CRCS, and fully available from the electronic health records [3]. We regard this as a MRCT with a binary outcome. Our key contribution is to provide efficient estimation for such a trial in the presence of noncompliance.

1.1 | Background and Significance

Noncompliance to treatment assignment is widespread in MRCTs including the e-assist trial. It decomposes the population of the e-assist trial into compliers and never takers based on the actual treatment assignment and receipt, and the hypothetical alternative assignment and receipt [4]: (i) experimental compliers who accessed e-assist; (ii) control compliers who would have participated in the intervention had they been assigned to e-assist; and (iii) never takers who would have avoided e-assist no matter what. We extend this approach by allowing the decomposition to occur within each and every site. We regard such site-specific decomposition as essential if different sites served different populations of patients or those sites varied in their effectiveness in implementing the treatment.

In the presence of noncompliance, researchers often estimate the per-protocol (PP) effect of treatment assignment after dropping observed non-compliers and as-treated (AT) effect of treatment received in addition to the intent-to-treat (ITT) effect of treatment assignment. Both PP and AT analyses fail to preserve the benefits of randomization and, thus, produce biased estimates of the treatment effect [5]. The ITT effect reveals the impact of assignment to the e-assist intervention, but, under noncompliance, is biased for the effect of the treatment received [6, 7]. Of interest in the study with noncompliance is the average treatment effect among compliers, also known as the complier average causal effect (CACE) [8]. With compliance as a partially observed pretreatment covariate whose missing pattern is explained by random treatment assignment in this paper, the CACE estimate will be causal and unbiased [7].

Little and Yau [4] conceived of noncompliance as a missing data problem and estimated the CACE of a job training intervention on a depression score with compliance assumed missing at random (MAR [9]). Small et al. [10] extended the MAR framework to a hierarchical model for estimation of the average CACE of an encouragement intervention on a binary depression outcome. They estimated the effect of encouragement as an instrumental variable (IV) on the adherence of patients to prescribed depression treatments by a logistic regression at the first stage. At the second stage, they estimated a logistic mixed model for the longitudinal outcome that depended on the fitted adherence and compliance, estimating time-specific fixed CACEs. Reardon, Unlu, and Bloom [11] used a theoretical random coefficient IV model where the site-specific effects of treatment assignment on a mediator and of the mediator on an outcome varied over sites and covaried with each other. They derived the IV estimators of the expected mediator effect, more accurate than the two-stage fixed-coefficient IV estimator, and compared bias in the estimated mediator effects by different IV approaches. Miratrix et al. [12] estimated the bounds of the CACE of a principal stratum [13], tightening them by pretreatment covariate stratification and estimating their uncertainty by the bootstrap. Zhou et al. [14] conducted a meta-analysis of studies comparing the rates of a cesarean section outcome among women in labor between epidural analgesia intervention and control (no or other analgesia treatment). They estimated the CACE of epidural analgesia on the cesarean delivery outcome by a logistic random intercept model where studies were conceived as clusters.

Other researchers have estimated the variance as well as mean of site-specific treatment effects in a MRCT. Raudenbush, Reardon, and Nomi [15] estimated hierarchical models for the school-specific impacts of an algebra program intervention on participation and math achievement separately, combining the estimates to produce the mean and variance of the school-specific CACE on math achievement of ninth graders. This approach assumes independence between school-specific compliance and CACE. Walters [16] analyzed Head Start (HS) experimental data where children aged 3–4 were randomly assigned or denied to attend a HS childcare center. Walters estimated a bivariate random coefficients model for the center-specific causal effects of assignment on a binary HS center participation and of the participation on a continuous cognitive skills outcome by maximum likelihood (ML). Of interest were the means, variances and covariance of the effects. This approach, however, assumes that the within-center latent variable underlying participation and potential outcomes are multivariate normal, a strong and uncheckable assumption [5]. Bloom et al. [17] estimated a theoretical hierarchical model with random intercepts, random ITT effects and lower-level error variances differing in experimental and control arms by a hybrid random coefficients model with fixed site-specific intercepts and the random treatment coefficient to estimate the mean and variance of site-specific ITT effects. Yuan, Feller, and Miratrix [18] estimated site-specific CACEs under conditional independence between site-specific compliance and CACE given pretreatment covariates, estimating the variance of the CACE by the bootstrap. They estimated nonparametric site-specific CACE and complier proportion to fit a linear regression for the CACE outcome given the proportion.

In cluster randomized trials (CRTs), Jo, Asparouhov, and Muthén [19] analyzed a behavioral outcome of children nested within classrooms. While classrooms were randomly assigned to intervention or control conditions at the cluster level, compliance of parents to the assignment was partial at the child level. The authors estimated the joint distribution of a mixture of the continuous outcome and binary noncompliance by ML via the EM algorithm, estimating the CACE and assuming uncorrelated compliance and outcome. They illustrated how ignoring noncompliance may result in drastically reduced power to detect the ITT effect. Jo et al. [20] extended the approach to the joint distribution of the outcome having compliance-specific random effects at lower and cluster levels and the binary compliance, and examined the effects of the three intra-cluster correlation coefficients (ICCs), complier-specific and noncomplier-specific ICCs of the continuous outcome and the ICC of the compliance, on variance associated with the CACE and power to detect it in various CRT settings by Monte Carlo simulations. Using nonparametric randomization inference, Small, Ten Have, and Rosenbaum [21] tested the magnitude of the effects of a randomly assigned depression care manager to primary care practices on group-randomized patients. Compliance of individuals to the assignment was partial, and covariate adjustment was used without requiring a model assumption. These approaches, however, cannot be easily extended to a multisite randomized trial where site-specific CACEs on a binary outcome vary randomly over sites.

1.2 | Contribution of the Current Article

In this article, we extend the ML estimation of Little and Yau [4] to two-level data that regards partially observed compliance as a missing data problem. Unlike the IV approaches [10, 11] and univariate estimations [15, 17, 18], we estimate the bivariate distribution of binary outcome and compliance by ML. While Walters [16] estimated the joint distribution of a continuous outcome and a binary participation, assuming a normally distributed latent variable underlying participation, we estimate the joint distribution of a binary outcome and a binary compliance by ML where site-specific CACE and log odds for experimental compliers, control compliers and never takers are jointly distributed random coefficients.

Our key contribution is to a multisite trial where each site generates its own causal effect. Useful statistics are the mean log odds for never takers and compliers assigned to each arm that may be compared with the mean CACE to reveal how effective the e-assist intervention is. Also useful are the variance and covariances of the physician-specific CACE. A negative correlation between CACE and log odds for control compliers, for example, implies that e-assist is effective among physicians whose complier patients would have completed CRCS poorly in the absence of the e-assist intervention, thereby promoting equality in CRCS. We will also estimate the site-specific CACEs and confidence intervals to identify physicians whose patients produce significant CACEs. Learning from the characteristics of the patients and physicians may shed light on who the beneficiaries of the effective e-assist treatment are.

The literature provides little guidance on how to study the joint distribution of physician-specific CACE and effects of control compliers, e-assist compliers, and never takers. Because compliance is missing for each control but observed for every experimental patient, the missing pattern depends on treatment assignment and, thus, the MAR assumption is met [4]. We estimate a bivariate random coefficients model for the binary outcome and compliance by ML where the physician-specific coefficients for never takers, control compliers, and e-assist compliers are expressed to be randomly varying over physicians. The difference in the random coefficients of e-assist and control compliers is defined as the CACE.

We fully observe the screening outcome Y_{ij}, treatment assignment T_{ij} and receipt D_{ij}, and the compliance of each experimental patient: patient i assigned to e-assist ($T_{ij} = 1$) is a complier ($C_{ij} = 1$) if the patient accessed it ($D_{ij} = 1$) and a never taker ($C_{ij} = 0$) otherwise ($D_{ij} = 0$). However, the compliance of every control is missing: although patient i assigned to control ($T_{ij} = 0$) is not allowed to access e-assist ($D_{ij} = 0$), we do not know whether the patient would have participated had that patient been assigned to e-assist. If compliance were also fully observed, the likelihood of parameters θ would have been

$$L(\theta) = \prod_j \int \prod_i f(Y_{ij}|C_{ij}, \nu_j) f(C_{ij}|\nu_j) \phi(\nu_j) d\nu_j$$

for patient i nested within physician j. Here $f(Y_{ij}|C_{ij}, \nu_j)$ is the conditional distribution of the screening outcome Y_{ij} given compliance C_{ij} and physician-specific random effects ν_j, $f(C_{ij}|\nu_j)$ is the distribution of compliance given ν_j, and $\nu_j \sim \phi(\nu_j)$ is multivariate normal. We, however, observe compliance partially and, thus, express our likelihood

$$L(\theta) = \prod_j \int \left\{ \prod_{\{i:1,1\}} f(Y_{ij}, 1|\nu_j) \prod_{\{i:1,0\}} f(Y_{ij}, 0|\nu_j) \right.$$
$$\left. \prod_{\{i:0,0\}} [f(Y_{ij}, 0|\nu_j) + f(Y_{ij}, 1|\nu_j)] \right\}$$
$$\phi(\nu_j) d\nu_j$$

for $f(Y_{ij}, C_{ij}|\nu_j) = f(Y_{ij}|C_{ij}, \nu_j) f(C_{ij}|\nu_j)$ and the set $\{i : T_{ij} = t, D_{ij} = d\} = \{i : t, d\}$ of patients assigned to treatment t who received treatment d within physician j for $t, d, = 0, 1$. We then approximate the integral numerically by adaptive Gauss–Hermite quadrature (AGHQ) [22–25], estimating θ by ML via AGHQ.

Our theoretical model for potential outcomes, ITT effects and CACE are consistent with well-established literature [4, 8, 26, 27]. Our key innovation is to model unique ITT effects and CACE within each and every cluster, defined in our case as physicians or physician practices. Our estimation theory is maximum likelihood estimation of fixed parameters and empirical Bayes estimates of cluster-specific ITT and CACE effects within the framework of a generalized linear mixed effects model or a hierarchical generalized linear model [28–32]. These general approach of maximum likelihood plus empirical Bayes is well established in the literature (e.g., Carl Morris' 1983 review). However, the problem of estimating the model from incomplete data under the MAR assumption is particularly challenging as it requires a doubly iterative approach using adaptive Gauss–Hermite quadrature (AGHQ). The MAR assumption itself is not problematic in this case given random assignment as we have explained. The key insight is to factor the likelihood into one piece that conditions on compliance C and another piece that is the distribution of C. This requires an extra step in numerical integration based on AGHQ. The identification of multiple covariance components in this setting is quite challenging, particularly when the within-cluster sample sizes are small and variable as they are here. However, the shared random effects model enables a more parsimonious representation that, if reasonable, significantly assists us in obtaining meaningful estimates of those covariance parameters. We also provide empirically meaningful simulation results in the particular context of our model.

The rest of the article is organized as follows. The next section explains the causal assumptions that will relate our estimators to the causal estimands. Section 3 presents the model and likelihood, and Section 4 shows how we estimate the model by ML. Section 5 illustrates analysis of the e-assist experimental data. Section 6 assesses our estimators by simulation, and the last section discusses the limitations and extensions.

2 | Causal Assumptions and Model

Let $T_{ij} = 1$ if patient i is assigned to e-assist and 0 otherwise within physician j for $i = 1, \ldots, n_j$ and $j = 1, \ldots, J$. Denote all assignments $\mathbf{T} = (T_{11}, T_{21}, \ldots, T_{n_J J})$, a potential participation $D_{ij}(\mathbf{T}) = 1$ if the patient participates in e-assist and 0 otherwise, and all participations $\mathbf{D} = (D_{11}(\mathbf{T}), D_{21}(\mathbf{T}), \ldots, D_{n_J J}(\mathbf{T}))$. A potential outcome of the patient is $Y_{ij}(\mathbf{T}, \mathbf{D})$ equal to 1 if the patient completes colorectal cancer screening and 0 otherwise. To make causal inference using the Rubin's causal model framework [33], we make the following assumptions [4, 7, 8, 26, 27]:

Assumption 1. (no interference between patients). The potential outcome and participation of a patient depend on the treatment status of oneself, but of no other patient. This assumption, also known as the stable unit treatment value assumption (SUTVA) [34], implies $D_{ij}(\mathbf{T}) = D_{ij}(T_{ij})$ and $Y_{ij}(\mathbf{T}, \mathbf{D}) = Y_{ij}(T_{ij}, D_{ij}(T_{ij}))$;

Assumption 2. (monotonicity). The impact of treatment assignment on participation is monotone, i.e., $D_{ij}(1) \geq D_{ij}(0)$;

Assumption 3. (random assignment). Treatment assignment is random so that the potential outcomes and participations are independent of the treatment assignment;

Assumption 4. (nonzero average causal effect of T_{ij} on D_{ij}). The expected difference in the proportions of patients receiving e-assist between those assigned to e-assist and control is nonzero. That is, $E[D_{ij}(1) - D_{ij}(0)] \neq 0$;

Assumption 5. (exclusion restriction). A potential outcome given participation does not depend on treatment assignment, that is, treatment assignment affects a potential outcome only through its impact on participation:

$$Y_{ij}(t) = Y_{ij}(t, D_{ij}(t)) = Y_{ij}(D_{ij}(t)) \text{ for } t = 0, 1.$$

Because experimental compliers accessed e-assist through their personal portal accounts, they were unlikely to share access to e-assist with other patients; neither controls nor experimental never takers were allowed the access. Furthermore, each eligible patient was pre-randomized to one of the treatment conditions before the trial began. Once the trial started, the patient was assigned to the condition. Consequently, the physician could not influence the treatment assignment in any possible way. Thus, the *no interference between patients* assumption seems reasonable. Yet, a scenario can be made to violate this assumption. If some experimental patients communicated with other experimental compliers or their physicians and were encouraged to or discouraged not to participate in e-assist, this assumption could be violated. However, patients received an email about random assignment and responded to it following a physician office visit. Consequently, it was highly unlikely for them to communicate about e-assist with each other during a physician office visit.

Because the trial does not allow controls to access e-assist, that is, $D_{ij}(0) = 0$, we overrule the existence of "defiers" who would take the opposite of what is assigned no matter what ($D_{ij}(1) = 0$ and $D_{ij}(0) = 1$) and "always takers" who would find a way to participate in e-assist no matter what ($D_{ij}(1) = 1$ and $D_{ij}(0) = 1$) [8]. Therefore, the *monotonicity*assumption $D_{ij}(1) \geq D_{ij}(0) = 0$ reasonably implies that a patient is either "complier" ($C_{ij} = 1$) if $D_{ij}(1) = 1$ and $D_{ij}(0) = 0$ or "never taker" ($C_{ij} = 0$) if $D_{ij}(1) = 0$ and $D_{ij}(0) = 0$; experimental patients are observed compliers if they participated in e-assist and never takers otherwise while the compliance status of every control is unknown as depicted in Table 1.

The *random assignment* assumption is reasonable by randomization of e-assist assignment. The *nonzero average causal effect of T_{ij} on D_{ij}* assumption is also reasonable because 78.34% of experimental patients accessed e-assist while no control was allowed the access; see Table 2.

The *exclusion restriction* assumption implies a zero causal effect for a never taker whose $D_{ij}(1) = D_{ij}(0) = 0$: $Y_{ij}(D_{ij}(1)) - Y_{ij}(D_{ij}(0)) = 0$. It reasonably states that a never taker cannot be influenced by treatment assignment. In other words, we assume that e-assist has no effect if no patient in the population uses e-assist after randomization. However, if control patients learned about the contents of e-assist from experimental compliers which influenced the outcome, then the *exclusion restriction* assumption could be violated. Again, because an experimental patient learned about random assignment and decided to or not to comply and access the e-assist by email following a physician's office visit, it is unlikely that such a discussion occurred between patients in the office. We therefore believe that these assumptions are quite plausible in the e-assist trial.

Each patient has two potential outcomes $Y_{ij}(t)$ for $t = 0, 1$, and one causal effect $Y_{ij}(1) - Y_{ij}(0)$. We write a potential outcome

$$\begin{aligned} &Y_{ij}(t) \sim \text{Bernoulli}(\pi_{tij}), \\ &\text{logit}(\pi_{tij}) = (1 - C_{ij})(\alpha_n + u_{nj}) + C_{ij}(\alpha_{ct} + u_{ctj}),\ t = 0, 1 \end{aligned} \tag{1}$$

TABLE 1 | One-sided noncompliance of the e-assist trial.

		D_{ij}	
		1	**0**
T_{ij}	1	c	n
	0	—	c or n

Abbreviations: c = complier; n = never taker.

for the physician-specific CRCS rates $\pi_{0ij} = \pi_{1ij} = [1 + exp\{-(\alpha_n + u_{nj})\}]^{-1}$ for never takers by the *exclusion restriction* assumption, $\pi_{0ij} = [1 + exp\{-(\alpha_{c0} + u_{c0j})\}]^{-1}$ for control compliers and $\pi_{1ij} = [1 + exp\{-(\alpha_{c1} + u_{c1j})\}]^{-1}$ for experimental compliers. Here $(u_{nj}, u_{c0j}, u_{c1j})$ are physician-specific random deviations from the mean log odds $(\alpha_n, \alpha_{c0}, \alpha_{c1})$ for never takers, control compliers and experimental compliers, respectively. We assume $u_{sj} \sim N(0, \tau_{ss})$ and $cov(u_{sj}, u_{s'j}) = \tau_{ss'}$ for $s, s' = n, c0, c1$. Of interest to us are a physician-specific $CACE_j = \alpha_{c1} + u_{c1j} - (\alpha_{c0} + u_{c0j})$, the mean CACE $\alpha_{c1} - \alpha_{c0}$ and how much CACE_j varies over physicians and covaries with $(u_{nj}, u_{c0j}, u_{c1j})$.

3 | Observed Model and Likelihood

Because we observe T_{ij}, $D_{ij} = T_{ij}D_{ij}(1) + (1 - T_{ij})D_{ij}(0)$ and $Y_{ij} = T_{ij}Y_{ij}(1) + (1 - T_{ij})Y_{ij}(0)$, model (1) implies the random coefficients model of interest to us

$$
\begin{aligned}
f(Y_{ij}|C_{ij}, u_j) &\sim \text{Bernoulli}(\pi_{yij}), \\
\eta_{yij} &= \text{logit}(\pi_{yij}) = A_{ij}^T\alpha + B_{ij}^T u_j
\end{aligned}
\tag{2}
$$

where $A_{ij}^T = \begin{pmatrix} 1 - C_{ij} & C_{ij}(1 - T_{ij}) & C_{ij}T_{ij} \end{pmatrix}$ is a vector of covariates having main and interactive fixed effects $\alpha = \begin{pmatrix} \alpha_n & \alpha_{c0} & \alpha_{c1} \end{pmatrix}^T$ and $B_{ij}^T = \begin{pmatrix} 1 - C_{ij} & C_{ij}(1 - T_{ij}) & C_{ij}T_{ij} \end{pmatrix}$ has main and interactive random effects $u_j = \begin{pmatrix} u_{nj} \\ u_{c0j} \\ u_{c1j} \end{pmatrix} \sim N\left(0, \tau = \begin{pmatrix} \tau_{nn} & \tau_{nc0} & \tau_{nc1} \\ \tau_{nc0} & \tau_{c0c0} & \tau_{c0c1} \\ \tau_{nc1} & \tau_{c0c1} & \tau_{c1c1} \end{pmatrix}\right)$. Extra pretreatment covariates W_{ij} may be added to A_{ij}, and their effects α_w may be added to α. The CACE of physician j is

$$
\begin{aligned}
CACE_j &= (\alpha_{c1} + u_{c1j}) - (\alpha_{c0} + u_{c0j}) \\
&\sim N(\alpha_{c1} - \alpha_{c0}, \tau_{c0c0} + \tau_{c1c1} - 2\tau_{c0c1})
\end{aligned}
\tag{3}
$$

Compliance, however, is partially observed. To handle missing compliance efficiently, we jointly estimate a hierarchical model for compliance

$$
f(C_{ij}|b_j) \sim \text{Bernoulli}(\pi_{cij}), \quad \eta_{cij} = \text{logit}(\pi_{cij}) = X_{ij}^T\gamma + Z_{ij}^T b_j \tag{4}
$$

where X_{ij} and Z_{ij} are vectors of pretreatment covariates having fixed effects γ and random effects $b_j \sim N(0, \Delta)$, respectively. The parameters are $\theta = (\alpha, \gamma, \tau, \Delta)$.

To express the likelihood, let $\nu_j = [u_j^T \; b_j^T]^T$ be p random effects having a probability density function (pdf) $\phi(\nu_j) = \phi(u_j|0, \tau)\phi(b_j|0, \Delta)$ for the pdf $\phi(\cdot|\mu, \sigma^2)$ of $N(\mu, \sigma^2)$. We denote the compliances C_{Ej} of all experimental patients ($< n_j$ patients) and all outcomes $Y_j = (Y_{1j}, \ldots, Y_{n_j j})$ within physician j to express the likelihood function of θ

$$
L(\theta) = \prod_j L_j, \quad L_j = \int f(Y_j, C_{Ej}|\nu_j)\phi(\nu_j)d\nu_j \tag{5}
$$

where

$$
\begin{aligned}
f(Y_j, C_{Ej}|\nu_j) = &\prod_{\{i: T_{ij}=1, D_{ij}=0\}} f(Y_{ij}|0, u_j)(1 - \pi_{cij}) \\
&\times \prod_{\{i: T_{ij}=1, D_{ij}=1\}} f(Y_{ij}|1, u_j)\pi_{cij} \\
&\times \prod_{\{i: T_{ij}=0, D_{ij}=0\}} [f(Y_{ij}|0, u_j)(1 - \pi_{cij}) + f(Y_{ij}|1, u_j)\pi_{cij}]
\end{aligned}
\tag{6}
$$

The first and second products decompose experimental patients into observed never takers and compliers, respectively, while the last product consists of controls whose compliances are all missing.

The integral L_j with respect to p dimensional ν_j is intractable. We choose to approximate L_j numerically by AGHQ. We therefore express the integrand $h(\nu_j) = f(Y_j, C_{Ej}|\nu_j)\phi(\nu_j)$ as a function of ν_j and calculate $\tilde{\nu}_j = E(\nu_j|Y_j, C_{Ej})$, $V_{\nu j} = Var(\nu_j|Y_j, C_{Ej})$ and a lower triangular matrix $L_{\nu j}$ by Cholesky decomposition $2V_{\nu j} = L_{\nu j}L_{\nu j}^T$ given θ estimated from the previous iteration. We then standardize ν_j by $x_{\nu j} = L_{\nu j}^{-1}(\nu_j - \tilde{\nu}_j)$. For each dimension of ν_j, we use Q representative values or abscissas $A = (a_1, \ldots, a_Q)$ of the standardized random effect and their weights $W = (w_1, \ldots, w_Q)$ proportional to the standardized pdf. Evaluation of L_j is then to sum the weighted $h(\nu_j)$ over the p^Q cells or abscissas of ν_j. Specifically, at a p-dimensional cell $(i_1, \ldots, i_p)$, we denote the abscissas $A_{i_1,\ldots,i_p} = [a_{i_1} \ldots a_{i_p}]^T$, weights $w_{i_1}, \ldots, w_{i_p}$ and adapted values $z_j = L_{\upsilon j}A_{i_1,\ldots,i_p} + \tilde{\upsilon}_j$ of ν_j to approximate

$$
\begin{aligned}
L_j &= \int \frac{\phi(\nu_j|\tilde{\upsilon}_j, V_j)}{\phi(\nu_j|\tilde{\upsilon}_j, V_j)} h(\nu_j)d\nu_j \\
&= |L_{\upsilon j}| \int e^{-x_{\upsilon j}^T x_{\upsilon j}} e^{x_{\upsilon j}^T x_{\upsilon j}} h(L_{\upsilon j}x_{\upsilon j} + \tilde{\upsilon}_j)dx_{\upsilon j}, \quad \nu_j = L_{\upsilon j}x_{\upsilon j} + \tilde{\upsilon}_j \\
&\approx |L_{\upsilon j}| \sum_{i_p=1}^{Q} \cdots \sum_{i_1=1}^{Q} w_{i_1} \ldots w_{i_p} e^{A_{i_1 \ldots i_p}^T A_{i_1 \ldots i_p}} h(z_j)
\end{aligned}
\tag{7}
$$

3.1 | Shared Random Effects

Imbens and Rubin [26] estimated a CACE by constraining the variance of the potential outcome for a never taker equal to that for a control complier and the variance of the potential outcome for an always taker equal to that for an experimental complier in a single-site trial, thereby allowing observed non-compliers to inform the variances of compliers whose compliance statuses are completely missing. We relax the constraints by shared random effects via a factor analytic structure, enabling all random effects to differ in variances and have pairwise covariances.

This approach also helps us overcome other difficulties we may confront. Random effects may be near the boundary of the parameter space with an almost zero variance or high correlations to slow the convergence. Furthermore, small cluster sizes with 10.7 patients per physician on average may constrain the estimability of multiple random effects. To test the hypothesis or improve the estimability, we apply shared random effects by factor loadings to model (2), extending the method of Miyazaki and Frank [35] to discrete outcomes. The reduced dimension of ν_j will also result in efficient computation by AGHQ.

Let $u'_j \sim N(0, \tau')$ be an r-by-1 vector ($r \leq 3$) of shared random effects having a variance covariance matrix τ' such that $u_j = \Lambda u'_j$ and $\tau = \Lambda \tau' \Lambda^T$ for a $3 \times r$ factor loading matrix Λ. Integral (5) is now with respect to $\nu_j = (u'_j, b_j)$ of a reduced dimension. With $r = 3$, Λ is a 3-by-3 identity matrix to imply $u_j = u'_j$ and $\tau = \tau'$. We consider two likely scenarios with u'_j and τ' of a dimension $r = 1$ or 2 based on our preliminary analysis.

We consider the first likely scenario of $r = 1$ where never takers, control compliers and e-assist compliers share a single random effect. Preliminary analysis implies that the random effect of observed never takers has a comparatively large variance to be away from zero and shared. Therefore, $u_j = \Lambda u'_j \sim N(0, \tau = \Lambda \tau' \Lambda^T)$ for

$$\Lambda = \begin{pmatrix} 1 \\ \lambda_1 \\ \lambda_2 \end{pmatrix}, \quad u'_j = u_{nj}, \quad \tau' = \tau_{nn}, \quad \tau = \tau_{nn} \begin{pmatrix} 1 & \lambda_1 & \lambda_2 \\ & \lambda_1^2 & \lambda_1 \lambda_2 \\ & & \lambda_2^2 \end{pmatrix} \tag{8}$$

having three parameters, less than six parameters of the unconstrained τ.

The next likely scenario is $r = 2$. Because every control of each physician has missing compliance, observed experimental never takers and compliers share their random effects with control compliers such that $u_j = \Lambda u'_j \sim N(0, \tau = \Lambda \tau' \Lambda^T)$ for

$$\begin{gathered} \Lambda = \begin{pmatrix} 1 & 0 \\ \lambda_1 & \lambda_2 \\ 0 & 1 \end{pmatrix}, \; u'_j = \begin{pmatrix} u_{nj} \\ u_{c1j} \end{pmatrix}, \; \tau' = \begin{pmatrix} \tau_{nn} & \tau_{nc1} \\ \tau_{nc1} & \tau_{c1c1} \end{pmatrix} \\ \tau = \begin{pmatrix} \tau_{nn} & \lambda_1 \tau_{nn} + \lambda_2 \tau_{nc1} & \tau_{nc1} \\ & \lambda_1^2 \tau_{nn} + 2\lambda_1 \lambda_2 \tau_{nc1} + \lambda_2^2 \tau_{c1c1} & \lambda_1 \tau_{nc1} + \lambda_2 \tau_{c1c1} \\ & & \tau_{c1c1} \end{pmatrix} \end{gathered} \tag{9}$$

having 5 parameters. We may reduce the dimension of symmetric τ further by setting one of the factor loadings to zero (e.g., $\lambda_2 = 0$).

4 | Maximum Likelihood Estimation

We view (Y_j, C_j, u'_j, b_j) as complete data (CD) and (Y_j, C_{Ej}) observed to estimate variance components (τ', Δ) by the EM algorithm [36] and fixed effects and factor loadings $\theta_f = [\alpha^T \ \gamma^T \ \lambda_1 \ \lambda_2]^T$ by the Newton Raphson (NR) method. We derive the expected CD ML estimators and NR estimators given random effects ν_j and, then, integrate them with respect to ν_j numerically by AGHQ. The EM estimators will converge stably to ML, and the NR estimators will converge faster than all EM estimators for efficient computation [37, 38]. Shin and Raudenbush [39] called such ν_j "provisionally known random effects."

To derive the NR estimators, let φ_τ and φ_Δ be the vectors of distinct elements in τ' and Δ, respectively, and write $\theta = [\theta_f^T \ \varphi^T]^T$ for $\varphi = [\varphi_\tau^T \ \varphi_\Delta^T]^T$. Denote the loglikelihood $l = \sum_j l_j$, score $S = \sum_j S_j$ and observed information $\mathcal{I} = -H = -\sum H_j$ for $l_j = \ln L_j$, $S_j = \frac{\partial l_j}{\partial \theta} = L_j^{-1} \frac{\partial L_j}{\partial \theta}$ and $H_j = L_j^{-1} \frac{\partial^2 L_j}{\partial \theta^T \partial \theta} - S_j S_j^T$. We estimate $\frac{\partial L_j}{\partial \theta}$ and $\frac{\partial^2 L_j}{\partial \theta^T \partial \theta}$ by AGHQ in Equation (7) where $h(\nu_j) = \frac{\partial}{\partial \theta} f(Y_j, C_{Ej} | \nu_j) \phi(\nu_j)$ and $h(\nu_j) = \frac{\partial^2}{\partial \theta^T \partial \theta} f(Y_j, C_{Ej} | \nu_j) \phi(\nu_j)$, respectively. At iteration $(m+1)$, we estimate score $S = [S_{\theta_f}^T \ S_\varphi^T]^T$ and observed information $\mathcal{I}$, consisting of on-diagonal submatrices $\mathcal{I}_{\theta_f \theta_f}$ and $\mathcal{I}_{\varphi\varphi}$ and off-diagonal $\mathcal{I}_{\theta_f \varphi}$, to obtain the NR estimators $\hat{\theta}^{(m+1)} = \hat{\theta}^{(m)} + \mathcal{I}^{-1} S$ given iteration-m estimates $\hat{\theta}^{(m)}$.

Estimation of the NR estimates failed to converge with a singular $\mathcal{I}$ due to comparatively small random effects and small cluster sizes in our e-assist data analysis. Convergence rates improve with higher variances of random effects or larger cluster sizes in our simulation. Because small random effects imply comparatively small $var(\hat{\varphi})$ and small $cov(\hat{\theta}_f, \hat{\varphi})$ from $\mathcal{I}^{-1}$, we adopted the NR estimator $\hat{\theta}_f^{(m+1)} = \hat{\theta}_f^{(m)} + \mathcal{I}_{\theta_f \theta_f}^{-1} S_{\theta_f}$ assuming $cov(\hat{\theta}_f, \hat{\varphi}) \approx 0$, and chose to estimate φ by the EM algorithm which would converge stably to ML.

The CD model $f(Y_{ij} | C_{ij}, u'_j) f(C_{ij} | b_j) \phi(\nu_j)$ in Equations (2) and (4) implies the CD ML estimators $\hat{\tau}' = \frac{\sum_j u'_j u'^T_j}{J}$ and $\hat{\Delta} = \frac{\sum_j b_j b_j^T}{J}$. We let $g(\nu_j) = \nu_j$ or $\nu_j \nu_j^T$ to find the E components

$$E[g(\nu_j) | Y_j, C_{Ej}] = \int g(\nu_j) f(\nu_j | Y_j, C_{Ej}) d\nu_j \tag{10}$$

where the nonstandard pdf by the Bayes theorem is

$$f(\nu_j | Y_j, C_{Ej}) = L_j^{-1} f(Y_j, C_{Ej} | \nu_j) \phi(\nu_j) \tag{11}$$

for L_j approximated by AGHQ and others in closed form. We evaluate the integral (10) by AGHQ in Equation (7) for $h(\nu_j) = L_j^{-1} g(\nu_j) f(Y_j, C_{Ej} | \nu_j) \phi(\nu_j)$.

At convergence, we estimate standard errors by $var(\hat{\theta}) = \mathcal{I}^{-1}$. See Appendix A of the Supporting Information for derivation of the estimators given random effects. To find the initial values, we fit the compliance model (4) given $T_{ij} = 1$, impute missing compliance by predictive mean matching from the fitted compliance model [40], and fit the outcome model (2) given the imputation. Specifically, to impute the missing compliance of each control, the predictive mean matching assigns to the control the observed compliance of the experimental patient who was closest in predicted mean within a physician. The initial factor loadings are then set at $\lambda_1 = \frac{\tau_{nc0}}{\tau_{nn}}$ and $\lambda_2 = \frac{\tau_{nc1}}{\tau_{nn}}$ if $r = 1$, and $\lambda_1 = \tau_{nc0}/\tau_{nn} - (\tau_{nc1}/\tau_{nn})(\tau_{nc1} - \tau_{nc0}\tau_{nc1}/\tau_{nn})/(\tau_{c1c1} - \tau_{nc1}^2/\tau_{nn})$ and $\lambda_2 = (\tau_{c0c1} - \tau_{nc0}\tau_{nc1}/\tau_{nn})/(\tau_{c1c1} - \tau_{nc1}^2/\tau_{nn})$ if $r = 2$.

In the next two sections, we estimate each model by our program that takes convergence to ML to be the square root of summed squared differences in the estimates between two consecutive iterations $< 10^{-4}$. We test a hypothesis at a level $\alpha = 0.05$. The program may be downloaded from https://github.com/sunx63/CACE.

5 | Data Analysis

We analyze 1825 participants nested within 170 physicians from the e-assist trial where 919 were assigned to e-assist and 906 to control. The CRCS outcome is fully observed from the electronic health records. We summarize data for analysis according to the observed treatment assignment (T) and receipt (D), compliance (C) and outcome (Y) in Table 2. Out of the 1825 participants who all had patient portal accounts, 65% completed CRCS above the national average 54% at the time of Lafata et al. [2]. Published findings imply that portal account holders may differ in measured and unmeasured ways from those without a portal account [41]. Although 65.88% of compliers and 58.29% of never takers assigned to e-assist completed CRCS, the rates of those assigned to control are unknown due to missing compliance.

A popular instrumental variable (IV) approach is to estimate the ITT effect on the outcome (0.01, standard error (SE) = 0.02) and participation (0.78, SE = 0.01) and find the CACE as their ratio (0.02, SE = 0.03) in terms of risk differences [7]. The IV estimation is not an appropriate analysis of our model (2) because random assignment occurred within each physician and compliance rates may vary from physician to physician.

We consider pretreatment patient covariates and physician's age in Table 3. Marital status and CRCS order remained unbalanced across treatment arms after randomization unfortunately. To test if marital status and CRCS order have significant effects on the outcome, we estimate the joint distribution in Equations (2) and (4) as the full model where W_{ij} comprises marital status and CRCS order in Equation (2), and X_{ij} consists of female indicator, Charlson comorbidity score [42], CRCS order and patient's age, and $Z_{ij} = 1$ in Equation (4) based on our preliminary analysis. The reduced model drops W_{ij} from Equation (2) of the full model. To see the impact of eight patients with missing marital status on our estimates, we fitted the reduced model with $N = 1825$ and, then, with $N = 1817$ patients after dropping the eight patients to find estimates barely changing and statistical inferences remaining the same between the two analyses (see Table 1 in Appendix B of the Supporting Information). Therefore, the impact of the eight missing marital statuses on the estimates is trivial. Next, given $N = 1817$ patients, we find that the full model is a better fit than the reduced one with the likelihood ratio test statistic 17.85 and p-value < 0.001. Table 4 shows the estimated reduced ($r = 3$) and full ($r = 3$) models in columns two and three, respectively.

TABLE 2 | Patients by treatment assignment and receipt, compliance, and outcome ($N = 1825$).

T	D	C	Y	# patients
0	0	Missing	0	322
0	0	Missing	1	584
1	0	n	0	83
1	0	n	1	116
1	1	c	0	232
1	1	c	1	488

Abbreviations: n = never taker; c = complier.

Given $N = 1817$, we now compare the full model with two reduced ones having shared random effects and other elements unchanged as explained in Section 3.1. The full model ($r = 3$) has six parameters in τ. We estimated one reduced model having three parameters in τ for $r = 1$ in Equation (8) where the random effect of never takers is shared with others, and the other having four parameters in τ for $r = 2$ in Equation (9) where we set $\lambda_2 = 0$ to have the random effect of never takers shared with control compliers. The sharing will inform the random effect of control compliers whose compliance statuses are completely missing. The AIC statistics are 3294.55, 3291.69, and 3288.28 for $r = 3$, 2 and 1, respectively [43]. Therefore, we do not find evidence that larger models are better fits than the reduced one for $r = 1$.

Although the effects of physician covariates are also of interest, the trial focused on patient outcomes and characteristics and, thus, collected only four physician characteristics: age, gender, specialty and location. Our analysis reveals the insignificant effects of the covariates except the small effect size of age that is statistically significant due to a small standard error (effect = 0.01, SE = 0.006, p-value = 0.034) and that produced the estimated odds ratio 1.01. The age effect remained both small and statistically insignificant after age was dichotomized into categories

TABLE 3 | Summary of patient characteristics and physician's age.

	e-assist (n = 919)	Control (n = 906)	p
Continuous, mean (sd)			
Patient's age	59.82 (7.29)	59.55 (7.22)	0.427
Physician's age (missing = 123)	51.40 (11.09)	51.71 (10.89)	0.561
Charlson comorbidity score	1.13 (1.81)	1.08 (1.79)	0.385
Discrete, n (%)			
Female	560 (60.94%)	580 (64.02%)	0.190
Marital status (missing = 8)			0.016
Married	626 (68.49%)	569 (63.01%)	
Single	288 (31.51%)	334 (36.99%)	
CRCS order			0.003
Colonoscopy	766 (83.35%)	705 (77.82%)	
Stool test only	153 (16.65%)	201 (22.19%)	

Note: Kruskal–Wallis tests were used for continuous variables, and Pearson's Chi-squared tests for discrete variables.

TABLE 4 | Estimated Models 2 and 4 from the e-assist trial data of $N = 1817$ patients.

	Reduced model ($r = 3$)		Full model ($r = 3$)		Reduced model ($r = 1$)	
Parameter	**Estimate (SE)**	p	**Estimate (SE)**	p	**Estimate (SE)**	p
α_n	0.35 (0.16)	0.013	0.17 (0.20)	0.195	0.17 (0.20)	0.200
α_{c0}	0.66 (0.10)	< 0.001	0.51 (0.16)	< 0.001	0.51 (0.16)	< 0.001
α_{c1}	0.78 (0.10)	< 0.001	0.60 (0.16)	< 0.001	0.60 (0.16)	< 0.001
$\alpha_{c1} - \alpha_{c0}$	0.12 (0.13)	0.195	0.09 (0.13)	0.246	0.10 (0.14)	0.242
$\alpha_{\text{marital status}}$	—	—	0.44 (0.11)	< 0.001	0.44 (0.11)	< 0.001
$\alpha_{\text{CRCS order}}$	—	—	−0.14 (0.13)	0.141	−0.14 (0.13)	0.142
γ_0	1.02 (0.23)	< 0.001	1.01 (0.23)	< 0.001	1.01 (0.23)	< 0.001
γ_{female}	0.22 (0.17)	0.10	0.22 (0.17)	0.095	0.22 (0.17)	0.100
γ_{CCS}	−0.04 (0.04)	0.169	−0.04 (0.04)	0.170	−0.04 (0.04)	0.180
$\gamma_{\text{CRCS order}}$	0.26 (0.21)	0.11	0.27 (0.21)	0.102	0.27 (0.21)	0.101
γ_{age}	−0.02 (0.01)	0.080	−0.02 (0.01)	0.081	−0.02 (0.01)	0.077
λ_1	—	—	—	—	0.30 (0.45)	0.249
λ_2	—	—	—	—	0.99 (0.57)	0.041
τ_{nn}	0.22		0.21		0.28	
τ_{nc0}	0.12		0.11		0.08	
τ_{nc1}	0.23		0.23		0.27	
τ_{c0c0}	0.06		0.06		0.03	
τ_{c0c1}	0.12		0.12		0.08	
τ_{c1c1}	0.25		0.24		0.27	
Δ	0.05		0.06		0.05	
log ML	−1639.25		−1630.27		−1630.14	
AIC	3308.50		3294.55		3288.28	

Note: CCS is Charlson comorbidity score.

such as age greater than the mean age 51.5 or not (odds ratio 1.21), and age > 60 or not (odds ratio 1.16) because physician's age and its missing pattern were balanced across the two treatment arms. We show the numerical summary of physician age in Table 3 that is missing for 21 physicians and, thus, their 123 patients. Because these covariates are all balanced across the treatment arms and because our method has yet to be extended to handle missing cluster-level covariates efficiently, we do not include the covariates in our analysis.

The number of n_j patients varies widely ranging 1 to 44 across physicians. To examine the impact of physicians with a few patients on our inferences, we dropped 45 physicians having $n_j < 5$, re-estimated the shared parameter model with $r = 1$ using 1732 patients within 125 physicians and compared the resulting estimates to those from full data in Table 2 in Appendix B of the Supporting Information. To facilitate the comparison, we copied the estimates from the last column of Table 4 to the last column of Table 2 in Appendix B. The estimates from the subset of the e-assist data are close to those from the full e-assist data and, thus, the resulting statistical inferences are identical across the two analyses.

We now interpret the estimated model for $r = 1$ in the last column of Table 4. The estimated fixed effects and standard errors are practically identical to those of the full model. Marital status is positively, but CRCS order is not significantly associated with the outcome, ceteris paribus. Controlling for the covariates, the mean log odds for never takers, control compliers and e-assist compliers are estimated to be 0.17, 0.51, and 0.60, respectively. The means of compliers are statistically significant, but the implied mean CACE is modest 0.10 (SE = 0.14, odds ratio (OR) = 1.11) with the p-value 0.242. This may be because the trial targeted a population of eligible patients having an online portal account who completed colorectal cancer screening (CRCS), higher than the national average at the time of Lafata et al. [2]. Therefore, the trial participants had comparatively little room for improved screening by e-assist to result in the modest effect.

The estimated factor loadings are positive. The estimates of τ, computed after convergence according to Equation (8), differ somewhat only in magnitude from those of the full model. The estimated variances are $\tau_{nn} = 0.28$ (SE = 0.29), $\tau_{c0c0} = 0.03$ (SE = 0.06) and $\tau_{c1c1} = 0.27$ (SE = 0.15). The random effects of e-assist compliers seem to have at least a non-negligible variance 0.27 above the standard error 0.15. The covariance $cov(u_{c1j}, u_{c1j} - u_{c0j}) = \tau_{c1c1} - \tau_{c0c1} = 0.19$ (SE = 0.15) with CACE_j is positive to imply that e-assist may be effective among physicians whose e-assist compliers completed CRCS well. The characteristics of the physicians and their e-assist complier patients may inform us of who the beneficiaries of the e-assist treatment are.

Estimation of the reduced model ($r = 1$) also produces the estimated CACEs of all physicians having 1 to 44 patients. Figure 1 graphs the mean CACE and each CACE_j with their confidence intervals (CIs). The solid horizontal line is the mean CACE (0.10) with the dashed lines drawing a 95% CI for it: $(-0.17, 0.36)$. The CI for the mean CACE is constructed by $(\hat{\alpha}_{c1} - \hat{\alpha}_{c0}) \pm 1.96 \times \sqrt{var(\hat{\alpha}_{c1}) + var(\hat{\alpha}_{c0}) - 2cov(\hat{\alpha}_{c1}, \hat{\alpha}_{c0})}$. Each dot plots a physician-specific CACE_j with its solid vertical line drawing a 95% CI for the CACE_j, where the CI for site-specific CACE is constructed by $(\hat{\alpha}_{c1} - \hat{\alpha}_{c0}) + E(u_{c1} - u_{c0}|Y_j, C_{Ej}) \pm 1.96 \times \sqrt{var(u_{c1} - u_{c0}|Y_j, C_{Ej})}$ in Equation (3). Although a number of physician-specific CACEs are located outside the CI for the mean CACE, all physician-specific CIs but one imply that nearly all physician practices produced CACEs that do not reliably differ from 0. Physician H9028 treated 22 eligible patients who produced the highest and significant CACE 0.64 with a 95% CI (0.10,1.19) and, thus, benefited from the e-assist intervention. The characteristics of the patients and physician may shed light on who the beneficiaries of the effective e-assist intervention are.

We compared our average CACE 0.10 (SE = 0.14, OR = 1.11) with the ITT effect, as-treated (AT) effect of treatment received, and per-protocol (PP) effect as the ITT effect of 1619 patients except observed never takers. The ITT effect 0.04 (SE = 0.10, OR = 1.04) is well known to be the product of the CACE and complier proportion and, thus, lower than the CACE. The AT effect 0.20 (SE = 0.11, OR = 1.22) is higher than the CACE as e-assist compliers achieve a higher CRCS rate than do others on average. The PP effect 0.14 (SE = 0.11, OR = 1.15) is higher than the CACE because observed never takers have a lower CRCS rate than do observed e-assist compliers on average.

Based on the estimates in the last column, we approximate $P(Y = 1|T = 1, C = 1) - P(Y = 1|T = 0, C = 1) \approx \frac{e^{0.6}}{1+e^{0.6}} - \frac{e^{0.51}}{1+e^{0.51}} = 0.02$, insignificant (SE = 0.03) and comparable to the IV estimate in terms of a risk difference. This is because our compliers seem highly health literate, thereby completing CRCS well above the national average at the time of Lafata et al. [2] regardless of whether e-assist is present or not. Nevertheless, the IV estimate is not an appropriate analysis of our model (2), and does not provide any information about variability in effects over physicians shown in Figure 1.

Our adaptive Gauss–Hermite computations used 7 abscissas to estimate the full model ($r = 3$) and reduced model ($r = 1$) in the last two columns of Table 4. Because estimation of the full and reduced models took long 9.10 and 0.15 h per iteration, respectively, we implemented parallel computation, assigning computation for each quarter of the $J = 170$ physicians per core and, thus, reducing the per-iteration time roughly by a factor of 4. Estimation of the full and reduced models converged to ML in 21 and 292 iterations, respectively. The computation time will improve as much as desired given the sufficient number of cores $\leq J$.

6 | Simulation Study

We evaluate our estimators by simulating the e-assist sample data and analysis from the previous section closely. We simulate two cases of the final model with $r = 1$ in the previous section: the large cluster sizes $n_j = 40$ patients and comparatively small cluster sizes of the e-assist sample ranging 1 to 44 patients (10.7 on average) within $J = 170$ physicians.

We simulate the large cluster sizes to validate our estimators and the correct execution of our program, and the e-assist cluster sizes to assess the accuracy and precision of our estimators in the previous section. For each case, we simulate data by the following steps:

1. $T_{ij} \sim \text{Bernoulli}(\pi_{Tj})$ by $\text{logit}(\pi_{Tj}) \sim N(0.2, 0.2)$ to target $\pi_{Tj} \approx 0.5$ on average;
2. $C_{ij} \sim \text{Bernoulli}(\pi_{cij})$ where $\text{logit}(\pi_{cij}) = 1 - 0.5X_{1ij} + 0.5X_{2ij} + b_j$ and $b_j \sim N(0, \Delta = 0.3)$ for $X_{1ij} \sim N(1, 1)$ and $X_{2ij} \sim Bin(1, 0.65)$ to simulate $\pi_{cij} \approx 0.7$ on average;

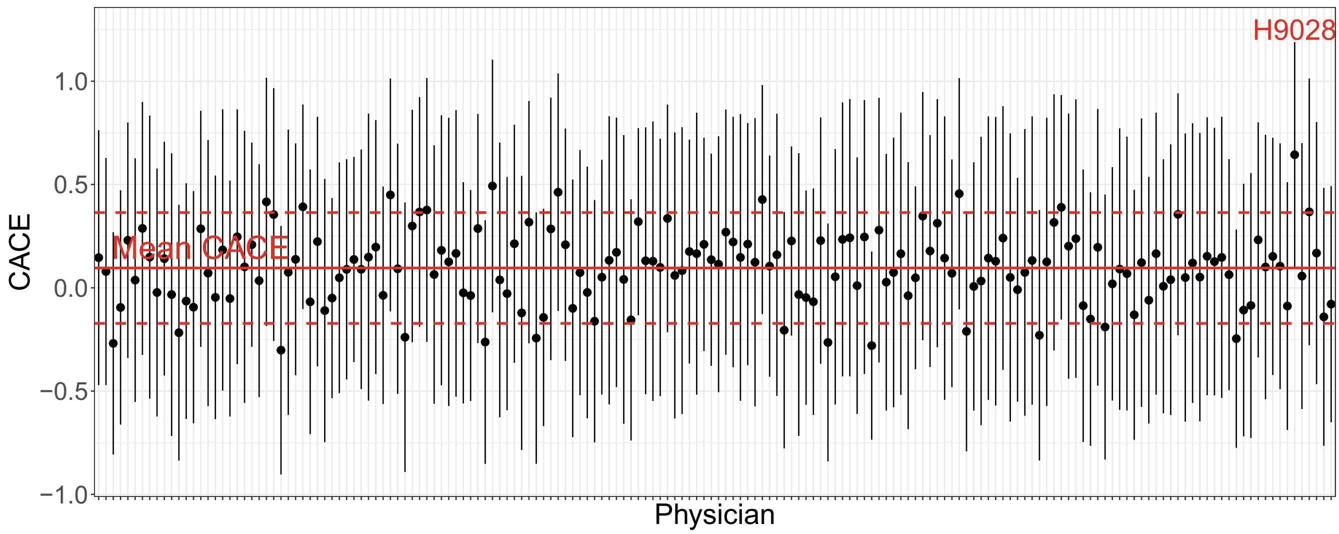

FIGURE 1 | Physician-specific CACEs ($r = 1$).

3. following Equation (1), $Y_{ij}(t) \sim \text{Bernoulli}(\pi_{tij})$ and $\text{logit}(\pi_{tij}) = 0.5(1 - C_{ij}) + 0.7C_{ij}(1 - t) + 1.2C_{ij}t - 0.5X_{1ij} + X_{2ij} + \Lambda u'_j$ to produce CACE $= 0.5$ and $\pi_{yij} \approx 0.69$ on average for $r = 1$, $u'_j \sim N(0, 0.5)$, $\lambda_1 = 0.7$ and $\lambda_2 = 1.1$ in Equation (8);
4. Set $D_{ij} = 1$ if $T_{ij} = C_{ij} = 1$ and 0 otherwise such that $P(D_{ij} = 1) \approx 0.39$ on average with about 50% of compliance missing.

We repeat simulation of data and estimation of $\theta = (\alpha, \gamma, \Lambda, \tau', \Delta)$ 500 times, using four abscissas by R [44]. Table 5 lists five columns: cluster sizes n_j, parameters, simulated true parameters values, gold standard (GS) univariate estimations of the simulated model (1) and compliance model (4) given all potential outcomes or $N = 2 \times 1825$ observations and fully known compliance by the lme4 package in R [45], and our ML estimation of the joint model in Equations (2) and (4) given observed data or $N = 1825$ observations and compliance MAR. The GS method estimates the full model (2) because the lme4 R package cannot estimate the shared random effects model. We converted our estimates back to the τ estimates according to Equation (8) for comparison because available software cannot estimate the shared random effect model. Each cell of the last two columns reports % bias (average standard error or ASE), the empirical estimate of the true standard error over simulated samples (ESE), and a 95% coverage probability.

With a large cluster size $n_j = 40$, both GS and our ML estimates have small biases up to −3.14% and 2.94%, respectively,

TABLE 5 | Simulated analysis for $r = 1$ and $J = 170$. Each cell lists %bias (average SE), empirical estimate of the true SE over simulated samples and a 95% coverage probability (CP).

			%bias (ASE), ESE, CP	
Sample size	**Parameter**	**Simulated**	**Complete data**	**Observed data**
$n_j = 40$	α_n	0.50	0.80 (0.07), 0.07, 0.95	1.45 (0.12), 0.10, 0.95
	α_{c0}	0.70	0.29 (0.06), 0.06, 0.94	0.94 (0.10), 0.10, 0.93
	α_{c1}	1.20	0.50 (0.08), 0.08, 0.93	0.27 (0.11), 0.09, 0.96
	α_1	−0.50	0.40 (0.02), 0.02, 0.95	0.34 (0.03), 0.03, 0.94
	α_2	1.00	0.10 (0.04), 0.04, 0.95	0.33 (0.06), 0.06, 0.95
	γ_0	1.00	−0.20 (0.07), 0.07, 0.96	−0.58 (0.09), 0.09, 0.93
	γ_1	−0.50	0.20 (0.03), 0.03, 0.97	−0.83 (0.04), 0.04, 0.94
	γ_2	0.50	−0.20 (0.06), 0.06, 0.94	0.67 (0.08), 0.08, 0.94
	τ_{nn}	0.50	−1.00 (0.09), 0.08, 0.92	0.50 (0.13), 0.13, 0.82
	τ_{nc0}	0.35	−3.14 (< 0.01), 0.05, 0.04	−1.18 (0.07), 0.06, 0.94
	τ_{nc1}	0.55	−2.72 (< 0.01), 0.07, 0.03	−0.99 (0.10), 0.10, 0.89
	τ_{c0c0}	0.24	0.82 (0.07), 0.05, 0.97	2.94 (0.09), 0.08, 0.92
	τ_{c0c1}	0.38	−2.34 (< 0.01), 0.06, 0.08	−0.54 (0.08), 0.08, 0.93
	τ_{c1c1}	0.61	−0.17 (5.11), 0.10, 0.93	−0.40 (0.12), 0.12, 0.93
	Δ	0.30	−2.33 (0.07), 0.05, 0.98	0.54 (0.06), 0.07, 0.93
	CACE	0.50	1.00 (0.06), 0.06, 0.94	−0.68 (0.10), 0.09, 0.95
n_j = e-assist	α_n	0.50	−0.20 (0.12), 0.12, 0.93	1.49 (0.18), 0.18, 0.94
	α_{c0}	0.70	0.29 (0.10), 0.11, 0.95	1.45 (0.18), 0.18, 0.95
	α_{c1}	1.20	0.17 (0.12), 0.13, 0.93	0.41 (0.16), 0.16, 0.94
	α_1	−0.50	−0.20 (0.04), 0.04, 0.94	0.87 (0.06), 0.06, 0.96
	α_2	1.00	0.20 (0.08), 0.08, 0.95	0.29 (0.12), 0.12, 0.95
	γ_0	1.00	−0.50 (0.12), 0.12, 0.93	−1.11 (0.16), 0.16, 0.95
	γ_1	−0.50	−0.40 (0.06), 0.06, 0.94	−1.02 (0.08), 0.08, 0.95
	γ_2	0.50	2.00 (0.11), 0.11, 0.94	1.93 (0.15), 0.15, 0.95
	τ_{nn}	0.50	0.40 (0.16), 0.14, 0.95	10.23 (0.30), 0.27, 0.87
	τ_{nc0}	0.35	−4.00 (0.01), 0.09, 0.12	−4.02 (0.14), 0.15, 0.94
	τ_{nc1}	0.55	−5.82 (0.12), 0.12, 0.15	0.52 (0.18), 0.19, 0.88
	τ_{c0c0}	0.24	10.20 (0.23), 0.09, 0.97	15.68 (0.19), 0.21, 0.88
	τ_{c0c1}	0.38	−2.86 (0.01), 0.09, 0.16	−2.01 (0.17), 0.18, 0.92
	τ_{c1c1}	0.61	1.16 (2.29), 0.16, 0.95	2.22 (0.25), 0.26, 0.93
	Δ	0.30	−3.00 (0.12), 0.09, 1.00	0.07 (0.14), 0.13, 0.93
	CACE	0.50	0.00 (0.11), 0.12, 0.94	−1.03 (0.19), 0.19, 0.95

most ASEs close to ESEs, and coverages near nominal 0.95. The GS covariance estimates, however, produce comparatively large biases, and ASEs lower than the ESE counterparts to result in poor coverages. The comparatively over-represented biases and under-represented ASEs seem to result from estimating the full τ when the true random effects are shared. Overall, our ASE and ESE estimates are comparatively a little inflated to reflect extra uncertainty due to small sample sizes and missing compliances.

With the small e-assist cluster sizes, GS and our biases of fixed effect estimates are again small up to 2%, but those of variance and covariance estimates are comparatively inflated up to 10.20% and 15.68% of τ_{c0c0}, respectively. Again most ASEs are close to ESEs with good coverages, but some GS covariance estimates have comparatively large biases, and small ASEs to result in poor coverages. Our ASE and ESE are somewhat inflated compared to the GS counterparts as before. Therefore, our estimators are reasonably accurate and precise for both sample sizes except some estimators of τ moderately biased for the e-assist cluster sizes.

Alternative analysis of the e-assist trial is to analyze the 1825 patients nested within 44 clinics, the smaller number of 44 clusters that may be more appropriate in many other multisite trials. We repeated the simulated analysis using $n_j = 40$ patients nested within each of $J = 44$ clinics near the e-assist sample size. The resulting estimates in Table 6 are largely comparable to those of the e-assist size reported in Table 5 although some variance estimates appear a little more accurate. Therefore, the estimates from 170 physicians and 44 clinics appear comparable.

The simulation results in Tables 5 and 6 also exhibit the consistency of our ML estimators. Our estimators become more accurate and precise as the cluster sizes increase from the e-assist size to a larger 40 within $J = 170$ clusters in Table 5 and, also, as the number of J clusters increases from 44 to 170 when the cluster sizes n_j is fixed at 40 in Tables 5 and 6.

7 | Discussion

We analyzed a binary colorectal cancer screening (CRCS) outcome from a multisite trial where patients were randomly assigned within physicians to e-assist, an online program that aimed to improve the CRCS rate, or control. Compliance to treatment assignment classified each experimental patient as a never taker or a complier, but was missing for all control patients. Of interest was the joint distribution of physician-specific complier average causal effects (CACEs) and log odds for never takers, and compliers assigned to each arm. Assuming compliance missing at random, we estimated a bivariate random coefficients model for the binary outcome and compliance by maximum likelihood (ML) where the random coefficients had the joint distribution. We expressed the likelihood and derived ML estimators given random effects, integrated them numerically with respect to the random effects by adaptive Gauss–Hermite quadrature (AGHQ), and assessed the estimators by simulation.

We estimated reduced models with the random effects of never takers, control compliers and experimental compliers shared by a factor analytic structure and tested their goodness of fits. We considered two scenarios: the random effect of never takers shared with others ($r = 1$) and the random effects of never takers and experimental compliers shared with control compliers ($r = 2$). The factor analytic structure enables the random effects of never takers, control compliers and experimental compliers to differ in variances and have covariances. One downside of $r = 1$ is that

TABLE 6 | Simulated analysis for $r = 1$ and $J = 44$. Each cell lists %bias (average SE), empirical estimate of the true SE over simulated samples and a 95% coverage.

			%bias (ASE), ESE, CP	
Sample size	**Parameter**	**Simulated**	**Complete data**	**Observed data**
$n_j = 40$	α_n	0.50	2.40 (0.14), 0.14, 0.95	−0.06 (0.20), 0.21, 0.94
	α_{c0}	0.70	1.86 (0.12), 0.11, 0.96	0.61 (0.18), 0.18, 0.96
	α_{c1}	1.20	1.42 (0.15), 0.16, 0.94	1.00 (0.19), 0.20, 0.92
	α_1	−0.50	0.20 (0.04), 0.04, 0.94	0.82 (0.06), 0.06, 0.97
	α_2	1.00	0.10 (0.08), 0.08, 0.95	1.06 (0.12), 0.11, 0.96
	γ_0	1.00	−0.10 (0.14), 0.13, 0.95	−0.32 (0.17), 0.17, 0.94
	γ_1	−0.50	0.20 (0.06), 0.06, 0.94	0.10 (0.08), 0.08, 0.95
	γ_2	0.50	0.20 (0.11), 0.11, 0.96	1.49 (0.15), 0.15, 0.95
	τ_{nn}	0.50	2.20 (0.18), 0.15, 0.90	6.11 (0.25), 0.26, 0.91
	τ_{nc0}	0.35	−2.86 (0.01), 0.10, 0.10	−4.43 (0.13), 0.13, 0.92
	τ_{nc1}	0.55	−3.27 (0.01), 0.15, 0.09	−0.23 (0.19), 0.20, 0.91
	τ_{c0c0}	0.24	6.94 (0.12), 0.09, 0.95	9.49 (0.18), 0.18, 0.90
	τ_{c0c1}	0.38	−2.34 (0.01), 0.11, 0.07	−0.84 (0.17), 0.17, 0.90
	τ_{c1c1}	0.61	0.83 (1.48), 0.19, 0.92	3.18 (0.25), 0.26, 0.90
	Δ	0.30	−3.33 (0.13), 0.10, 0.98	−3.57 (0.12), 0.13, 0.89
	CACE	0.50	0.80 (0.12), 0.12, 0.95	1.50 (0.19), 0.19, 0.94

the reduced parameters were able to identify the signs, but not magnitudes, of correlations in Equation (8).

Our approach estimated physician-specific CACEs with confidence intervals to reveal their variability and identify physicians whose patients produced significant CACEs. The characteristics of the patients and physicians may shed light on who the beneficiaries of the effective e-assist intervention are.

We estimated our bivariate generalized random coefficients model with four random coefficients varying across J clusters, which requires a relatively large number of clusters. Although, we simulated as low as $J = 44$ clusters to produce reasonably accurate and precise estimates, we do not recommend estimating the model with significantly fewer clusters than the simulated $J = 44$ unless the number of random coefficients is reduced.

Our ML estimation of the bivariate random coefficients model for binary outcome and compliance may be extended to the estimation of a joint exponential family distribution. Specifically, the binary outcome with the logit link in Equation (2) may be extended to a discrete or continuous outcome model with an appropriate link from an exponential family distribution. Likewise, the binary compliance with the logit link in Equation (4) may be extended to more principal strata from a multinomial distribution [27, 46].

We implemented parallel computation by four CPU cores to reduce the per-iteration time roughly by a factor of four in estimation. This approach may cut down the computation time further by adding more cores up to the number of J sites. Other than parallel computing, we have not attempted to optimize our computation. A valuable future research topic is to improve computation time further by reducing the number of iterations, for example, via the parameter expanded EM algorithm [47] or DAAREM [48].

Our current method is built to handle missing values of compliance. A valuable future research is to extend our approach to handling partially observed covariates or outcomes. This extension is also important because of the need to involve auxiliary variables to strengthen the MAR assumption [39, 49]. Partially observed variables added to the model will grow the dimension of random effects and burden the numerical integration by AGHQ. Numerical integration by multivariate Laplace approximation may result in more efficient computation [50].

Finally, we used the five standard causal assumptions from the literature given a fully observed outcome. With a partially observed outcome in the future research, however, additional assumptions may be required if the missing pattern of Y is nonignorable and depend, for example, on latent compliance strata [46, 51, 52].

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Additional supporting information can be found online in the Supporting Information section.